

EXECUTIVE BRIEF

Samples That Can Represent a Nation

What leaders should authorize, require, and refuse

Production Unit	PU-B07-C17
Decision status	Controlled learning asset
Audience	Executives, programme owners, statisticians, technical leads
Data	Synthetic examples only

Executive takeaway: Design agricultural samples whose target population, frame, stratification, selection probabilities, weights, and precision support national decisions.

Why it matters

A large sample is not nationally representative when coverage, selection probabilities, or weights are defective.

Illustrative application

A stratified sample preserves selection probabilities and calculates weights, nonresponse adjustments, and design-based precision.

Minimum approval conditions

- Named accountable owner and lawful purpose
- Defined evidence, provenance, reference period, population, and geography
- Decision-relevant validation and visible uncertainty
- Documented abstention, referral, override, and incident processes
- Accessibility, equity, security, retention, and retirement review

Non-negotiable boundary

The analytical system may inform review. It may not independently authorize benefits, sanctions, enforcement, clinical treatment, school exclusion, displacement, or another consequential action.